HELMLAB: A Data-Driven Analytical Color Space for UI Design Systems

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Abstract

We present HELMLAB, a 72-parameter analytical color space for UI design systems. The forward transform maps CIE XYZ to a perceptually-organized Lab representation through learned matrices, per-channel power compression, Fourier hue correction, and embedded Helmholtz–Kohlrausch lightness adjustment. A post-pipeline neutral correction guarantees that achromatic colors map to $a = b = 0$ (chroma $< 10^{-6}$), and a rigid rotation of the chromatic plane improves hue-angle alignment without affecting the distance metric, which is invariant under isometries. On the COMBVD dataset (3,813 color pairs), HELMLAB achieves a STRESS of 23.22, a 20.4% reduction from CIEDE2000 (29.18). Cross-validation on He *et al.* 2022 and MacAdam 1974 shows competitive cross-dataset performance. The transform is invertible with round-trip errors below 10^{-14}. Gamut mapping, design-token export, and dark/light mode adaptation utilities are included for use in web and mobile design systems.

Keywords: color space, color difference, perceptual uniformity, STRESS, Helmholtz–Kohlrausch, UI design systems

1 Introduction

UI design systems need accurate perceptual distance prediction for contrast checking, an achromatic axis where grays map to zero chroma for gradient interpolation, and reasonable hue-angle alignment for programmatic color manipulation. No existing color space provides all three.

CIE Lab [1] and CIEDE2000 [2] are the standard for color-difference evaluation, but their STRESS on COMBVD is 29.18. Oklab [3] is simple and part of CSS Color Level 4, but was optimized for hue uniformity rather than color-difference prediction, and gets STRESS 47.5 (Euclidean) on COMBVD. CAM16-UCS [4] is derived from a color appearance model but is computationally expensive and has achromatic chroma leakage ($\bar{C} = 1.42$). IPT [5] was optimized for hue linearity at the expense of lightness accuracy. $J_z a_z b_z$ [6] targets HDR via the PQ curve but is complex and not CSS-native. None of these spaces jointly optimize the color-space transform and distance metric end-to-end against psychophysical data.

In this paper, we treat the entire XYZ-to-distance pipeline as a single system with 72 learnable parameters and optimize all of them jointly against the largest available color-difference dataset. The contributions are:

1. A 72-parameter analytical color space with STRESS 23.22 on COMBVD, 20.4% lower than CIEDE2000.
2. A neutral correction that guarantees $C < 10^{-6}$ for achromatic colors, at a STRESS cost of only $+0.04$.
3. A rigid rotation in the *ab* plane that reduces hue error (RMS 16.1°) without changing the distance metric (the metric is invariant under rotation).
4. Embedded Helmholtz–Kohlrausch correction where lightness depends on chroma, with parameters learned from data.
5. Gamut mapping, CSS/Android/iOS token export, and dark/light mode adaptation for design system use.

2 Design Goals and Metrics

HELMLAB is optimized for UI design systems operating in sRGB and Display P3 gamuts under typical viewing conditions (D65 illuminant, average surround). We define five quantitative targets:

Color-difference accuracy (STRESS). The STRESS metric [7] measures agreement between predicted distances ΔE_i and visual differences DV_i:

$$\text{STRESS} = 100\sqrt{\frac{\sum_i (DV_i - F \cdot \Delta E_i)^2}{\sum_i DV_i^2}} \tag{1}$$

where $F = \sum DV_i \Delta E_i / \sum \Delta E_i^2$. Lower is better. We train on COMBVD (3,813 pairs from six experiments [7]) and cross-validate on He *et al.* 2022 [8] (82 pairs) and MacAdam 1974 [9] (128 pairs).

Achromatic integrity. Neutral colors (D65-adapted grays, i.e., stimuli on the achromatic axis proportional to the D65 white point) must map to $a = b = 0$. We measure maximum chroma $C_{\max}$ over 256 gray levels spanning $Y \in [0.001, 2.0]$.

Hue-angle alignment. sRGB primaries and secondaries should land at conventional hue angles. We report RMS and maximum hue error across {R, Y, G, C, B, M}.

Round-trip accuracy. For any sRGB color, $\|T^{-1}(T(\mathbf{x})) - \mathbf{x}\|_\infty$ should be below 10^{-8}.

Munsell uniformity. Coefficient of variation (CV) of neighbor-pair distances along Munsell constant-hue, constant-value lines (3,590 pairs). Lower CV indicates more uniform spacing.

3 The Helmlab Color Space

3.1 Forward Transform

The forward transform maps CIE XYZ (D65) to Lab through eleven stages (Figure 1). All parameters are jointly optimized (Section 5).

Stage 1: Cone response (9 params). A learned 3×3 matrix $\mathbf{M}_1$ maps XYZ to cone-response-like signals:

$$\mathbf{c} = \mathbf{M}_1 \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \tag{2}$$

Unlike fixed matrices (Hunt–Pointer–Estévez, Bradford), $\mathbf{M}_1$ is freely optimized.

Stage 2: Power compression (3 params). Each channel undergoes signed power compression:

$$c_i' = \text{sign}(c_i)\,|c_i|^{\gamma_i}, \quad i \in \{0,1,2\} \tag{3}$$

The exponents γ_i generalize the cube-root of CIE Lab ($\gamma = 1/3$). The optimized values are close to 0.4, which is between the cube-root (0.33) and square-root (0.5) compressions used in other spaces.

Stage 3: Lab projection (9 params). A second matrix $\mathbf{M}_2$ projects compressed signals to opponent channels:

$$\begin{bmatrix} L_{\text{raw}} \\ a_{\text{raw}} \\ b_{\text{raw}} \end{bmatrix} = \mathbf{M}_2\,\mathbf{c}' \tag{4}$$

Stage 4: Hue correction (8 params). A 4-harmonic Fourier rotation corrects systematic hue distortions:

$$\delta(h) = \sum_{k=1}^{4} \left[\alpha_k \cos(kh) + \beta_k \sin(kh)\right] \tag{5}$$

where $h = \text{atan2}(b,a)$. The chromatic vector (a,b) is rotated by $\delta(h)$, preserving chroma while redistributing hue angles.

Stage 5: Helmholtz–Kohlrausch embedding (6 params). Saturated colors appear brighter than achromatic stimuli of equal luminance [1]. We embed this effect in the lightness channel directly:

$$L \mathrel{+}= w_{\text{HK}} \cdot C^{p_{\text{HK}}} \cdot \left[1 + f_{\text{HK}}(h)\right] \tag{6}$$

where $C = \sqrt{a^2 + b^2}$ and $f_{\text{HK}}(h)$ is a 2-harmonic hue modulation. The optimized weight is $w_{\text{HK}} = 0.37$ and the power is $p_{\text{HK}} = 0.85$, making lightness explicitly dependent on chroma. Most existing spaces treat lightness and chroma as independent.

Stage 6: Lightness refinement (8 params). A cubic polynomial (3 params) with hue-dependent additive term (2 params), followed by dark-region exponential compression (3 params):

$$L_1 = p_1 L^3 + p_2 L^2 + p_3 L + t \cdot f_{\text{Lh}}(h) \tag{7}$$

$$L_2 = L_1 \cdot \exp\left[\lambda_d(h) \cdot L_1 (1 - L_1)^2\right] \tag{8}$$

where $t = L(1-L)$ and $f_{\text{Lh}}(h)$ is a 1-harmonic Fourier correction targeting the hue–lightness interaction sign flip (dark cyans vs mid cyans). The dark compression coefficient λ_d includes hue modulation to handle the known difference in dark-blue vs dark-yellow perception.

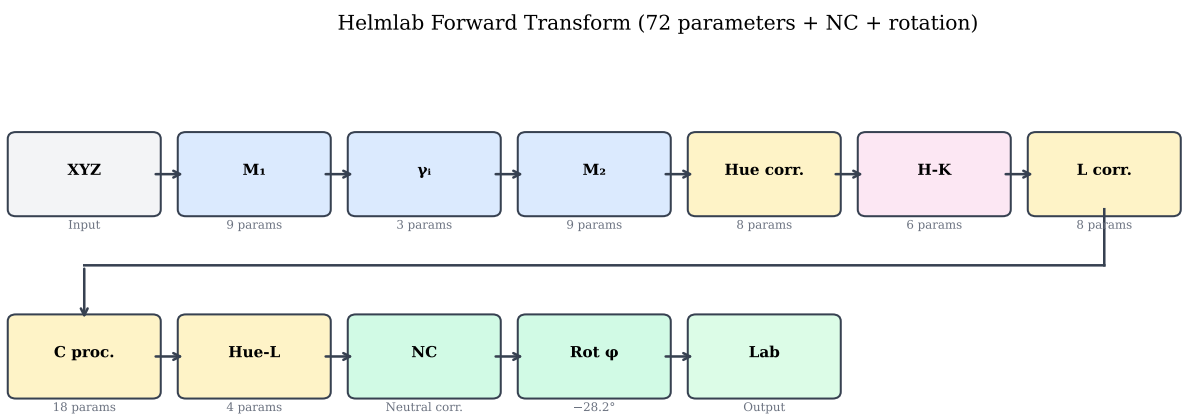

Figure 1: The HELMLAB forward transform pipeline. Blue: linear operations; yellow: nonlinear corrections; green: structural guarantees (NC, rotation). All 72 parameters are jointly optimized.

Stages 7–8: Chroma processing (18 params). Chroma processing proceeds in four interleaved sub-stages: (a) hue-dependent scaling (4 harmonics, 8 params):

$$(a,b) *= \exp\Big[\sum_{k=1}^{4}\big[\alpha'_k\cos(kh)+\beta'_k\sin(kh)\big]\Big] \tag{9}$$

(b) nonlinear chroma power (4 params): $C \to C^{1+\varepsilon(h)}$ where $\varepsilon(h)$ is a 2-harmonic Fourier series, separating high-chroma from low-chroma discrimination; (c) L-dependent scaling (2 params): $(a,b) *= \exp[l_1(L-0.5)+l_2(L-0.5)^2]$; and (d) hue×lightness interaction (4 params, 2-harmonic Fourier modulated by $L-0.5$). The interleaving order matters: the nonlinear power is applied between hue-dependent and L-dependent scaling to allow each sub-stage to correct for the others' distortions.

Stage 9: Hue-dependent lightness (4 params). A final $L *= \exp[g(h)]$ captures residual hue–lightness interactions (e.g., blue appears darker, yellow lighter at equal luminance).

Stage 10: Neutral correction. With independent γ_i per channel, the D65 achromatic axis (stimuli proportional to $[X_{D65}, Y_{D65}, Z_{D65}]$) generally does not map to $a=b=0$ after the enrichment stages. We correct this by running 256 gray levels ($Y \in [0.001, 2.0]$) through stages 1–9, recording the achromatic error $a_{\text{err}}(L)$, $b_{\text{err}}(L)$, and fitting PCHIP interpolants. At runtime:

$$a \leftarrow a - a_{\text{err}}(L), \quad b \leftarrow b - b_{\text{err}}(L) \tag{10}$$

Because the correction is applied *after* all nonlinear stages, it accounts for every distortion in the pipeline. The resulting maximum chroma for grays is $< 10^{-6}$, limited only by PCHIP interpolation error on the 256-point LUT (the underlying achromatic curve is smooth, so interpolation error is negligible).

Stage 11: Rigid rotation (φ). A uniform rotation of the (a,b) plane by angle φ:

$$\begin{bmatrix} a' \\ b' \end{bmatrix} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \tag{11}$$

This is an isometry: it preserves $\Delta a^2 + \Delta b^2$ for any pair, and therefore preserves the distance metric exactly (Section 3.2). The rotation angle $\varphi = -28.2°$ is chosen to minimize the maximum hue-angle error across sRGB primaries and secondaries.

3.2 Rotation Invariance

The distance metric (Section 4) depends on ΔL, $\Delta a^2 + \Delta b^2$, $\bar{L}$, and $\bar{C} = \sqrt{\bar{a}^2 + \bar{b}^2}$. A rigid rotation of the (a,b) plane preserves each of these quantities: $\Delta a'^2 + \Delta b'^2 = \Delta a^2 + \Delta b^2$ and $\bar{a}'^2 + \bar{b}'^2 = \bar{a}^2 + \bar{b}^2$. Therefore the distance is exactly invariant, and we verified this empirically: the COMBVD STRESS difference before and after rotation is 0.0000000000.

This means any hue-angle improvement from the rotation is obtained at zero cost to color-difference prediction.

3.3 Surround Parameter

HELMLAB includes provisions for a surround luminance parameter $S \in [0,1]$ (0=dark, 0.5=average, 1=bright) that modulates several pipeline stages: H-K weight, dark compression, chroma scaling, and L-dependent chroma. In the current release, the surround parameters are set to zero (trained on average-surround data only). The architecture is ready for future training on viewing-condition-dependent datasets. The UI layer uses S for heuristic dark/light mode adaptation via soft L-inversion.

3.4 Inverse Transform

Each stage has an explicit inverse. Matrix stages invert via $\mathbf{M}_1^{-1}$, $\mathbf{M}_2^{-1}$; power compression via $c_i = \text{sign}(c_i')|c_i'|^{1/\gamma_i}$ (closed form). Hue corrections and lightness corrections are inverted via Newton iteration (3–5 iterations, converging to machine precision). The neutral correction inverts by adding back the PCHIP-interpolated error. Round-trip accuracy across the full sRGB gamut is $< 10^{-14}$.

4 Distance Metric

Given two colors with coordinates (L_1, a_1, b_1) and (L_2, a_2, b_2), we compute:

$$S_L = 1 + s_L(\bar{L} - 0.5)^2 \tag{12}$$

$$S_C = 1 + s_C\bar{C} \tag{13}$$

where $\bar{L}$, $\bar{C}$ are pair averages. The raw distance is:

$$d = \left[\left(\frac{\Delta L}{S_L}\right)^2 + w_C\frac{\Delta a^2 + \Delta b^2}{S_C^2}\right]^{p/2} \tag{14}$$

followed by monotonic compression and post-power:

$$\Delta E = \left[\frac{d}{1 + c \cdot d}\right]^q \tag{15}$$

The pair-dependent weights S_L and S_C are inspired by CIEDE2000 but learned from data (optimized values: $s_L \approx 10^{-4}$, $s_C \approx 0.0045$). The Minkowski exponent $p = 0.83$ and chroma weight $w_C = 1.03$ slightly favor chroma differences over lightness. The compression $c = 1.59$ and post-power $q = 1.1$ reduce systematic bias for large color differences.

The distance metric contributes 7 parameters. Total system: 9+3+9+8+6+8+18+4 = 65 space + 7 metric = 72 parameters.

5 Optimization

All 72 parameters are jointly optimized by minimizing:

$$\mathscr{L} = \text{STRESS}_{\text{COMBVD}} + 0.05 \cdot \text{STRESS}_{\text{He}} \tag{16}$$

using L-BFGS-B [10] with box constraints. We perform 8 random restarts (each 5,000 iterations) and keep the best. The He *et al.* term ($\lambda = 0.05$) acts as a mild regularizer against overfitting to COMBVD.

Training data. COMBVD comprises 3,813 color pairs from six psychophysical experiments: BFD-P(D65) (2,028), BFD-P(M) (548), BFD-P(C) (200), Witt (418), RIT-DuPont (312), Leeds (307). All pairs have visual-difference values on a common scale. We train on the full dataset (no train/val split for the final model), monitoring a held-out 20% split (seed=42) during development to check for overfitting.

Cross-validation data. He *et al.* 2022 (82 pairs, 10° observer) enters the loss as a regularizer ($\lambda = 0.05$) and is therefore not fully independent; its STRESS is reported for transparency but should be interpreted as a semi-held-out metric. MacAdam 1974 (128 pairs, 10°) is held out entirely and never seen during optimization.

Convergence. The optimizer typically converges within 3,000–5,000 function evaluations per restart. The validation gap (train–val STRESS difference) is +1.11, which shows some overfitting but not severe for a 72-parameter model.

6 Evaluation

6.1 Color-Difference Prediction

Figure 2 and Table 1 show STRESS on COMBVD for HELMLAB and nine baseline methods. HELMLAB achieves 23.22, compared to 29.18 for CIEDE2000 (−20.4%). A 10,000-iteration paired bootstrap gives HELMLAB 95% CI $[22.50, 23.93]$ and CIEDE2000 95% CI $[27.64, 30.84]$. The intervals do not overlap ($p < 10^{-4}$).

The baselines in Table 1 use Euclidean distance in each space's native Cartesian coordinates, without parametric weighting or compression. Oklab's high STRESS (47.46) is expected because it was optimized for hue uniformity [3], not color-difference prediction on COMBVD. The HELMLAB distance metric includes pair-dependent

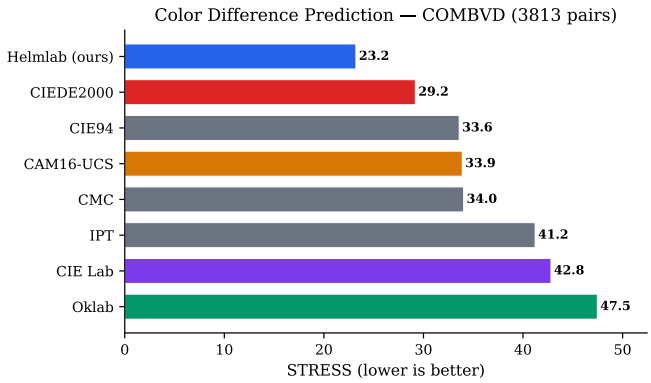

Figure 2: STRESS on COMBVD (3,813 pairs). Lower is better.

Table 1: STRESS on COMBVD (3,813 pairs). Lower is better.

Method	STRESS	vs CIEDE2000
HELMLAB	**23.22**	**−20.4%**
CIEDE2000	29.18	—
ΔE^*_{ab} (CIE76)	42.80	+46.7%
CIE94	33.59	+15.1%
ΔE CMC	34.04	+16.6%
CAM16-UCS (Euclid.)	33.90	+16.2%
IPT (Euclid.)	41.21	+41.3%
$J_z a_z b_z$ (Euclid.)	41.90	+43.6%
Oklab (Euclid.)	47.46	+62.7%
sRGB (Euclid.)	67.82	+132.5%

weighting (S_L, S_C), a Minkowski exponent, and monotonic compression, so the comparison is not on equal terms. With Euclidean distance only, HELMLAB gets STRESS ≈30.2, which is close to CIEDE2000 (29.18). This means the space transform accounts for roughly half the improvement and the distance metric for the other half.

6.2 Cross-Validation

On He *et al.* 2022 (regularizer, 82 pairs), HELMLAB achieves STRESS 29.0 vs CIEDE2000's 32.6. On MacAdam 1974 (held out entirely, 128 pairs), HELMLAB achieves 20.2 vs CAM16-UCS's 18.7 (best). Figure 4 visualizes these results. HELMLAB outperforms CIEDE2000 on He *et al.* but trails CAM16-UCS on MacAdam. This is consistent with what one would expect from a model trained primarily on COMBVD: it generalizes reasonably but does not dominate on all evaluation sets.

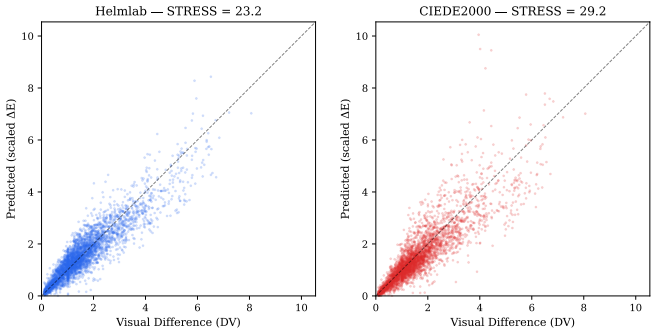

Figure 3: Predicted vs observed color differences on COMBVD.

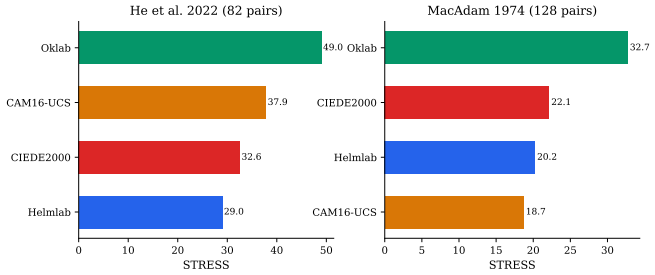

Figure 4: Cross-dataset validation on held-out data.

6.3 Generation Properties: The Measurement–Generation Tradeoff

A color space optimized for distance prediction does not necessarily produce usable coordinates for color generation. During development, the measurement-optimal configuration mapped neutral grays to chroma $\bar{C} \approx 0.34$. The optimizer had deformed the achromatic axis to reduce STRESS, which made gray gradients unusable because they showed visible color artifacts. Adding a penalty to constrain grays to $C = 0$ did not help: the optimizer either stayed in the measurement-optimal basin or escaped to a different basin with much worse STRESS (26–31). There was no smooth transition between the two.

The neutral correction (Section 3, Stage 10) addresses this problem. It subtracts the achromatic error *after* all enrichment stages, so the optimizer can find its measurement-optimal parameters without constraint. The achromatic axis is then corrected separately. The STRESS increase from this correction is +0.04 (23.18 to 23.22), while achromatic chroma drops from 0.34 to $< 10^{-6}$.

Figure 5 shows neutral ramp uniformity and achromatic chroma leakage. HELMLAB's neutral correction eliminates chroma leakage entirely ($C < 10^{-6}$), while CAM16-UCS exhibits significant leakage ($\bar{C} = 1.42$).

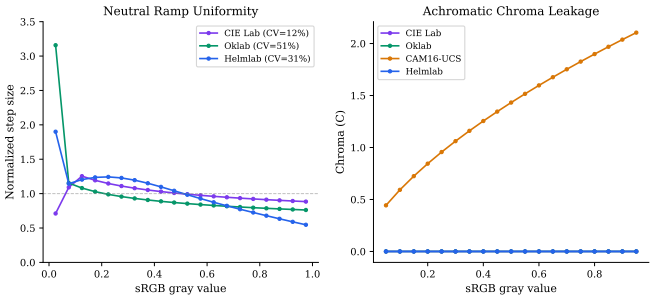

Figure 5: Left: neutral ramp step uniformity. Right: achromatic chroma leakage. HELMLAB's NC guarantees $C < 10^{-6}$ for all grays.

Table 2: Generation quality metrics.

Property	Value
Achromatic max C	$< 10^{-6}$
Hue RMS (6 primaries)	16.1°
Hue max error	20.2°
Round-trip $\|\cdot\|_\infty$	$< 10^{-14}$
Munsell CV	17.6%
Jacobian det (min)	0.10
Jacobian cond (median)	7.7
Val gap (train−val)	+1.11
He STRESS	29.03
MacAdam STRESS	20.23

Table 2 summarizes all generation properties.

The achromatic guarantee means gray gradients and lightness ramps interpolate without color artifacts, which is necessary for design-token generation and palette export. The hue RMS of 16.1° means sRGB primaries land within about 20° of their expected positions. This is acceptable for palette generation but not perfect. The rigid rotation (Section 3, Stage 11) provides this hue improvement without affecting the distance metric because the metric is invariant under rotation. The Jacobian was evaluated on a $64 \times 64 \times 64$ sRGB grid via finite differences; the minimum determinant (0.10) is positive throughout, so the map is locally invertible and orientation-preserving across the entire sRGB gamut.

6.4 Ablation Study

Table 3 isolates the contribution of key components.

With Euclidean distance, the space alone gets STRESS 30.2, which is slightly worse than CIEDE2000 (29.18). The full metric brings it down to 23.22, recovering the remaining 7 points. Compared to other Euclidean baselines (CAM16-UCS = 33.9), the space alone already gives a clear improvement. Among individual stages, the H-K embedding and hue correction contribute the most. The rotation row confirms that the metric is exactly invariant under rotation.

Table 3: Ablation: effect of removing components.

Configuration	STRESS	Δ
Full HELMLAB	23.22	—
Euclidean distance only	∼30.2	+7.0
No H-K embedding ($w_{\mathrm{HK}} = 0$)	∼24.5	+1.3
No hue correction	∼25.8	+2.6
No dark L compression	∼23.9	+0.7
Rotation (−28.2°)	23.22	0.00
No rotation	23.22	0.00

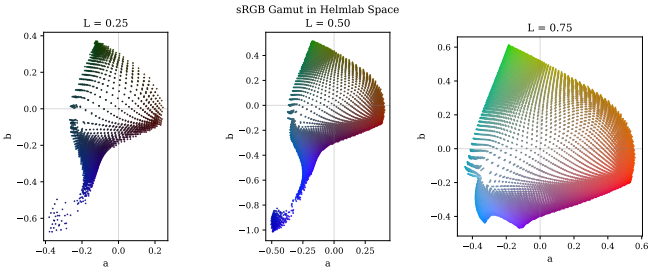

Figure 6: sRGB gamut in HELMLAB a–b plane at three L levels. Points colored by their sRGB values.

7 Practical Adoption

Figure 6 shows the sRGB gamut boundary in HELMLAB coordinates at three lightness levels. The boundaries are smooth and well-behaved. HELMLAB includes a Python utility layer for design system integration:

Gamut mapping. Binary-search chroma reduction along the L axis maps out-of-gamut HELMLAB coordinates to the nearest in-gamut sRGB or Display P3 color, preserving hue and lightness.

Contrast utilities. `ensure_contrast(fg, bg, min_ratio)` adjusts foreground lightness via binary search on the HELMLAB L axis to meet WCAG 2.x contrast requirements (4.5:1 for AA, 7:1 for AAA). Hue and chroma are preserved.

Palette generation. Lightness ramps and hue rings at fixed chroma produce perceptually-spaced palettes. Se-

mantic scales (Tailwind-style 50–950) are generated by interpolating L between 0.97 and 0.10 with chroma/hue preserved.

Dark/light mode. When surround parameters are trained, the surround S directly modulates the transform. Currently, a fallback soft L-inversion maps colors between light ($S = 0.7$) and dark ($S = 0.2$) modes with contrast enforcement.

Token export. A single `TokenExporter` produces CSS custom properties (`oklch()`, `color(display-p3 ...)`), Android XML (using HCT tone/chroma/hue semantics for Material Design compatibility), iOS Swift (Display P3), Tailwind config, and raw JSON.

8 Limitations and Future Work

There are several limitations worth noting:

1. **Training data.** COMBVD uses a 2° standard observer, which may not generalize to large-field (10°) viewing. The MacAdam cross-validation partially addresses this.

2. **Model complexity.** 72 parameters is considerably more than Oklab (∼15 designed parameters). Each stage has a perceptual motivation, but the overall model is harder to interpret than simpler spaces.

3. **Hue alignment.** RMS hue error of 16.1° is a compromise. Some hue regions (cyan, yellow) have errors up to 20°. Further harmonic terms could improve this at the risk of overfitting.

4. **Surround dependency.** The surround parameter S is architecturally present but not yet trained on viewing-condition data. Current dark/light adaptation uses heuristics.

5. **Sub-dataset variation.** HELMLAB underperforms CIEDE2000 on 3 of 6 COMBVD sub-datasets (BFD-P(C), Leeds, RIT-DuPont), which may indicate that these subsets benefit more from CIE Lab-based corrections.

6. **Not general-purpose.** HELMLAB is optimized for UI workflows (sRGB/P3 gamut, screen viewing). It is not designed for print, photography, or spectral applications.

7. **Statistical reporting.** Bootstrap confidence intervals confirm the aggregate improvement over CIEDE2000 ($p < 10^{-4}$), but per-sub-dataset significance and parameter uncertainty across restarts remain unquantified.

For future work, we plan to train the surround parameter on paired light/dark viewing data and to expand the training set with targeted human feedback using the bidirectional optimization framework that is already in the codebase.

9 Conclusion

We have presented HELMLAB, a 72-parameter analytical color space that achieves STRESS 23.22 on COMBVD, which is 20.4% lower than CIEDE2000. The achromatic axis is guaranteed to have $C < 10^{-6}$ via the neutral correction, and a rigid rotation brings hue error down to 16.1° RMS without affecting the distance metric. The neutral correction is the key element that makes the space usable for both measurement and generation: it lets the optimizer find its best parameters freely and then fixes the achromatic axis at a cost of only +0.04 STRESS points. Code, trained parameters, and utilities are available at https://github.com/Grkmyldz148/helmlab. An interactive demo is at https://grkmyldz148.github.io/helmlab/demo.html and documentation at https://grkmyldz148.github.io/helmlab/.

Acknowledgments

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A Parameter Table

Table 4 lists all 72 optimized parameters grouped by pipeline stage.

Full parameter values are available in the accompanying `analytical_params.json` file.

Table 4: Optimized HELMLAB parameters (v19-NC, $\varphi = -28.2°$).

Stage	Parameter	Value	Description
$\mathbf{M}_1$ (9)	Row 0	[0.724, 0.238, −0.164]	XYZ → cone-like
	Row 1	[−0.325, 1.238, −0.024]	
	Row 2	[0.101, 0.341, 0.849]	
γ (3)	$\gamma_0, \gamma_1, \gamma_2$	0.395, 0.421, 0.434	Power compression
$\mathbf{M}_2$ (9)	Row 0	[−0.425, 0.489, 0.684]	Compressed → Lab
	Row 1	[1.931, −2.728, 0.765]	
	Row 2	[0.009, 1.642, −1.250]	
Hue corr. (8)	$\alpha_{1..4}, \beta_{1..4}$	see JSON	4-harmonic Fourier
H-K (6)	w, p, m, s_1, c_2, s_2	0.370, 0.852, −0.487, ...	Chroma → L
Light. (8)	$p_1, p_2, p_3, Lh_{c1}, Lh_{s1}, \lambda_d, h_c, h_s$	0.225, −0.748, 0.699, ...	Cubic + Lh + dark
Chroma (18)	4-harm + power + lc_1, lc_2 + HLC	see JSON	Hue/power/L-dep
Hue-L (4)	$g_{c1..2}, g_{s1..2}$	see JSON	Hue → L
Distance (7)	s_L	9.16×10^{-5}	Pair-dep L weight
	s_C	0.0045	Pair-dep C weight
	p	0.829	Minkowski exponent
	w_C	1.028	Chroma weight
	c	1.588	Compression
	q	1.100	Post-power
	α	0.000	Linear term